Yu-Fan (Van) Lin

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EDUCATION

PhD in Computer Science **2026 – Present**
University at Albany, State University of New York, Albany, NY, USA
- Research Topic: Low-level Vision, Robust Perception, Multi-modal Learning, Video Restoration.
- Advisor: Prof. Ming-Ching Chang, Associate Professor (Computer Vision and Machine Learning Lab)

MSc in Miin Wu School of Computing **2024 – 2026**
National Cheng Kung University, Tainan, Taiwan
- Research Topic: Computer Vision, Image Processing, Low-level Vision, Multi-modal Learning.
- Advisor: Prof. Chih-Chung Hsu, Associate Professor

BSc in Applied Mathematics | GPA: 3.95/4.3 **2022 – 2024**
National Chung Hsing University, Taichung, Taiwan
- BS Thesis: Developing an Automatic Prohibited Items Detection Model for Airport Luggage X-ray Images Based on the Integration of Convolutional Neural Networks and Visual Transformers.
- Advisor: Prof. Chih-En Kuo, Associate Professor

BSc in Telecommunication Engineering (Transferred) | GPA: 3.96/4.3 **2020 – 2022**
National Kaohsiung University of Science and Technology, Kaohsiung, Taiwan
- Thesis title: An Intelligent Human-in-the-Loop Biometric Feedback System for Shooting Proficiency Enhancement.
- Advisor: Prof. Huang-Chu Huang, Professor

Affiliation / Experience

Research Collaboration, NVIDIA Taiwan Research **2024 – Present**
Project: Cybersecurity in Distributed Computing for Remote Sensing.
Mentor: Fu-En (Fred) Yang, Research Scientist

Industry Research Project, Taiwan High Speed Rail Corporation **2024 – 2025**
Automated Brake System Crack Detection: Design an automated brake system crack detection system, replacing manual inspection and reducing inspection time by 80%

AI Applications Engineer Intern, Information and Communications Research Laboratories (ICL), ITRI **2023 – 2024**
- Sole intern leading AI algorithm development for a large-scale automated retail deployment.
- Developed a non-standard viewpoint person ReID system, achieving >90% tracking accuracy.
- Built an automated out-of-stock detection system deployed across multiple application sites.

Plant Disease Analysis System, National Chung Hsing University **2023 – 2024**
Joint project with the Institute of Agronomy, NCHU, on plant disease analysis.

Awards and Honors

1. 3rd place, NTIRE Ambient Lighting Normalization Challenge, CVPR 2026.
2. Top 2%, PBVS Mars Landslide Segmentation Challenge, CVPR 2026.
3. Winner (1/16), SkiTB Visual Tracking Challenge, WACV 2026.
   Addressed long-term multi-object tracking in real-world ski racing videos, where **high-speed motion**, **frequent occlusions**, and **shot changes** cause severe trajectory fragmentation. Proposed an **anchor-frame-based bidirectional tracking** framework integrating RF-DETR and HiM2SAM, with bounding box refinement to maintain identity consistency across shot transitions.
4. 3rd place (3/136 & 3/274), Hyper-Object Challenge (Spectral Reconstruction & Super-Resolution), ICASSP 2026.
5. 4th place (4/418), Data-Centric Land Cover Classification Challenge, BMVC 2025.
   Addressed **multi-modal remote sensing fusion** across 14-channel SAR and MSI data, facing significant **domain gaps** and **severe class imbalance**. Proposed a **dual-stream architecture** combining LWGANet's remote-sensing-tailored Group Attention with DINOv3's semantic priors, integrated via a **Gated Fusion Module** for adaptive cross-modal feature injection.
6. Best paper award, CVGIP 2025.

7. Winner, Soccer Track Challenge, ACMMM 2025.
   Addressed **multi-object tracking under fisheye distortion** with extreme scale variation, uniform player appearance, and prolonged occlusions. Proposed GTATrack: combining Deep-EIoU for motion-agnostic online association and GTA-Link for global trajectory refinement, achieving HOTA=0.60 with significantly reduced false positives. doi: 10.1145/3728423.3759416

8. Winner, Multi-source COVID-19 Detection Challenge, ICCV 2025.
   Addressed **cross-institution domain shift** in multi-source chest CT classification, where varying scanners, protocols, and slice thickness cause severe generalization failure. Proposed input-space standardization combining SSFL++ (lung-centered spatial alignment) and KDS (density-based slice selection), reducing cross-source feature variance by 75% without requiring domain labels.

9. 5th place, Social Media Popularity Prediction Challenge, ACMMM 2025.
   Addressed **temporal distribution shift** in video popularity prediction, where models degrade as viral trends evolve. Proposed AMCFG: a TTA-inspired dual-anchoring framework combining multi-modal clustering with statistical anchors (cluster-level engagement patterns) and LLM-generated semantic anchors to discover temporally-invariant features.

10. Winner, TreeScope Lidar-based Tree Diameter Estimation Challenge, ICRA 2025.
    Addressed automatic tree diameter (DBH) estimation from outdoor LiDAR point clouds, facing challenges of **heavy noise, irregular geometry, and cross-scene generalization**. Proposed an end-to-end pipeline combining normal-guided density clustering for trunk extraction, dual geometric fitting (ellipse + cylinder), and XGBoost residual regression for robust diameter prediction.

11. Winner, GOOSE 2D Semantic Segmentation Challenge, ICRA 2025.
    Addressed outdoor semantic segmentation under **dramatic lighting variation** causing color shift and exposure issues. Proposed MaskDINO with Color Shift Estimation-and-Correction (CSEC) and Rotary Position Embedding for enhanced multi-resolution boundary features.

12. 3rd place (3/306, 0.9%), NTIRE Single Image Shadow Removal Challenge, CVPR 2025.
    Addressed high-resolution indoor shadow removal with **mixed hard/soft shadows and texture loss** in occluded regions. Proposed a frequency-enhanced multi-modal approach leveraging depth and surface normal guidance to restore fine details while maintaining structural consistency.

13. 6th place (6/263), NTIRE Single Image Reflection Removal Challenge, CVPR 2025.
    Addressed reflection removal with **high-frequency texture loss** in occluded regions. Proposed a multi-modal framework combining DINOv2 semantics, depth-normal guidance, and frequency-domain fusion (FreqFusion) with a two-stage robustness-refinement training strategy for robust high-frequency reconstruction.

14. 4th place (4/10), MaCVi MarineVision Restoration Challenge, WACV 2025.
    Tackled coupled restoration-detection for underwater imagery under **severe color distortion and depth-varying degradation**. Developed a LAB-space cross-attention network with physics-guided attenuation modeling, and a multi-scale fusion strategy balancing high-resolution structure preservation with enhanced color restoration.

15. Winner, Beyond Visible Spectrum: AI for Agriculture Challenge, ICPR 2024.
    Addressed Fusarium Head Blight detection from high-dimensional hyperspectral imagery without expensive hardware. Proposed a **self-supervised pipeline** with top-K band selection via clustering-guided feature importance, enabling accurate diagnosis with a lightweight classifier.

16. Top Performance Award, Social Media Popularity Prediction Challenge, ACMMM 2024.
    Identified overlooked **image-text semantic inconsistency** in social media posts, which increases with popularity. Proposed VLM feature adaptation with perplexity and similarity measures to align multi-modal representations with popularity distribution. doi: 10.1145/3664647.3689000

17. Presidential Award, Department of Applied Mathematics (Data Science and Computing Program), National Chung Hsing University, 2024.

18. Runner-up, Auto-WCEBleedGen Challenge V2, ICIP 2024.
    Addressed gastrointestinal bleeding detection under **severe class imbalance and high lesion-normal similarity**. Proposed a two-stage pipeline: classification-guided filtering followed by segmentation for accurate bleeding region localization.

19. Presidential Award, Department of Applied Mathematics (Data Science and Computing Program), National Chung Hsing University, 2023.

20. Runner-up, XRun! Motion Sensing Technology Innovation Competition, 2021.
    Designed an AI-powered shooting training system with real-time multi-sensor data fusion (hand pressure + motion). Developed an interactive GUI for performance feedback to assist targeted skill improvement.

21. Titanium Medal Award (Top 10%), Green Idea Invention and Design Fair, 2021.
    Developed a real-time tail light recognition system using object detection, integrated with a hardware-based sensing module for instant lane-change alerts to enhance driving safety.

22. Winner, Asian Robotic Athletic Competition, 2018.
Designed a light-sensing autonomous vehicle for line tracking with embedded symbol and digit recognition to perform arithmetic operations at designated checkpoints.

Scholarships

8th Foxconn Scholarship, 2024.
4th Foxconn Scholarship, 2020.
Awarded to only 30 students nationwide based on academic excellence and financial need. Beyond financial support, the program provides a Growth Camp for networking and personal development workshops, along with mentorship opportunities connecting recipients with peers and industry professionals.
Nan Shan Life Insurance Company Scholarship, 2024.

Professional Service

Conference Reviewer
AAAI Conference on Artificial Intelligence (AAAI), 2025.
IEEE International Conference on Multimedia and Expo (ICME), 2026.
Journal Reviewer
IEEE Transactions on Multimedia (TMM), 2025.
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2025.
International Journal of Pattern Recognition and Artificial Intelligence (IJPRAI), 2025

Journal Publications

1. C.-C. Hsu, C.-M. Lee, Y.-F. Lin, and L.-W. Kang, "S³RNet: Sparse Spatial–Spectral Representation with Hybrid Knowledge Distillation for Efficient Hyperspectral Image Pansharpening," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS), 2026.
2. C.-H. Cheng, C.-M. Lee, W.-T. Liao, Y.-F. Lin, Y.-C. Cheng, F.-E. Yang, Y.-C. F. Wang, et al., "PromptHSI: Universal Hyperspectral Image Restoration with Vision-Language Modulated Frequency Adaptation," IEEE Transactions on Geoscience and Remote Sensing (TGRS), 2026.

Refereed Conference Publications

1. R.-L. Jian, T.-Y. Chen, Y.-F. Lin, C.-M. Lee, F.-E. Yang, Y.-C. F. Wang, and C.-C. Hsu, "CANDLE: Illumination-Invariant Semantic Priors for Color Ambient Lighting Normalization," in IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
2. C.-M. Lee, Y.-F. Lin, J.-H. Jiang, Y.-J. Hsiao, C.-C. Hsu, and Y.-L. Liu, "ReflexSplit: Single Image Reflection Separation via Layer Fusion-Separation," in IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
3. C.-M. Lee, Y.-F. Lin, Y.-J. Hsiao, J.-H. Jiang, Y.-L. Liu, and C.-C. Hsu, "PhaSR: Generalized Image Shadow Removal with Physically Aligned Priors," in IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
4. C.-M. Lee, Y.-F. Lin, W.-H. Tsai, M.-C. Chang, X. Li, and C.-C. Hsu, "FracQuant: Rank-Adaptive Image Super-Resolution Network Quantization via Fractal Complexity Assessment," in International Conference on Machine Learning (ICML), CoLoRAI Workshop, 2026.
5. C.-M. Lee, Y.-H. Ho, Y.-F. Lin, J.-W. Lee, L.-W. Kang, and C.-C. Hsu, "HSSDCT: Factorized Spatial-Spectral Correlation for Hyperspectral Image Fusion," in IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Barcelona, Spain, 2026.
6. Z.-Y. Wu, C.-M. Lee, Y.-F. Lin, C.-C. Hsu, S.-Y. Chen, and L.-W. Kang, "HMS³Former: A Hyperspectral Image Restoration Model Based on Multi-Stage Spatial-Spectral Transformer and Endmember Attention," in IEEE International Conference on Image Processing (ICIP), 2026.
7. C.-M. Lee, B.-C. Qiu, T.-Y. Chen, Y.-F. Lin, F.-Y. Lin, J.-T. Tsai, I.-A. Tsai, and C.-C. Hsu, "Taming Domain Shift in Multi-Source CT-Scan Classification via Input-Space Standardization," in IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), 2025.
8. Y.-F. Lin, C.-M. Lee, and C.-C. Hsu, "DenseSR: Image Shadow Removal as Dense Prediction," in Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), Dublin, Ireland, Oct. 2025, pp. 7026–7035. doi:10.1145/3746027.3755664

9. R.-L. Jian, M.-C. Lo, C.-W. Huang, C.-M. Lee, Y.-F. Lin, and C.-C. Hsu, "GTATrack: Winner Solution to SoccerTrack 2025 with Deep-EIoU and Global Tracklet Association," in Proceedings of the 8th International ACM Workshop on Multimedia Content Analysis in Sports (MMSports '25), Dublin, Ireland, Oct. 2025. doi: 10.1145/3728423.3759416
10. C.-M. Lee, B.-C. Qiu, Y.-H. Wu, C.-J. Kang, J.-L. Chen, Y.-F. Lin, Y.-S. Chou, and C.-C. Hsu, "Anchoring Trends: Mitigating Social Media Popularity Prediction Drift via Feature Clustering and Expansion," in Proceedings of the 33rd ACM International Conference on Multimedia (ACMMM), 2025.
11. C.-M. Lee, Y.-F. Lin, Y.-H. Ho, L.-W. Kang, and C.-C. Hsu, "HyFusion: Enhanced Reception Field Transformer for Hyperspectral Image Fusion," in IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2025.
12. C.-M. Lee, Y.-F. Lin, L.-W. Kang, and C.-C. Hsu, "Robust Hyperspectral Image Pansharpening via Sparse Spatial-Spectral Representation," in IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2025.
13. C.-H. Cheng, Y.-F. Lin, C.-M. Lee, C.-C. Hsu, Y.-L. Shih, and C. Ting, "A Multi-Stage Deep Learning Pipeline for Crack Detection and Segmentation in High-Speed Rail Brake Discs," in IEEE International Conference on Consumer Technology (ICCT-Pacific), 2025.
14. C.-C. Hsu, C.-M. Lee, Y.-F. Lin, Y.-S. Chou, C.-Y. Jian, and C.-H. Tsai, "Revisiting Vision-Language Features Adaptation and Inconsistency for Social Media Popularity Prediction," in Proceedings of the 32nd ACM International Conference on Multimedia (ACMMM), Melbourne, Australia, 2024, pp. 11464–11469. doi: 10.1145/3664647.3689000

Preprints and Manuscripts Under Review

1. B.-C. Qiu, F.-Y. Lin, M.-H. Sun, Y.-F. Lin, C.-M. Lee, and C.-C. Hsu, "VISTA: Validation-Guided Integration of Spatial and Temporal Foundation Models with Anatomical Decoding for Rare-Pathology VCE Event Detection," arXiv:2605.22096, 2026.
2. C.-M. Lee, C.-J. Kang, C.-H. Cheng, Y.-F. Lin, Y.-S. Chou, C.-C. Hsu, F.-E. Yang, et al., "AuroraHSI: Degradation-Agnostic Hyperspectral Image Fusion Transformer via Mask-Guided Information Sharing and Compensation," 2025.
3. C.-M. Lee, Y.-F. Lin, L.-W. Kang, and C.-C. Hsu, "SSCP: Factorized Spatial-Spectral Correlation with Prompt-Guided Band Selection for Efficient Hyperspectral Image Fusion," 2025.
4. C.-M. Lee, C.-J. Kang, C.-H. Cheng, Y.-F. Lin, Y.-S. Chou, C.-C. Hsu, F.-E. Yang, et al., "RobustDGA: Generalized Hyperspectral Image Fusion via High-Frequency Information Sharing and Compensation," 2024.
5. Y.-F. Lin, C.-H. Cheng, B.-C. Qiu, C.-J. Kang, C.-M. Lee, and C.-C. Hsu, "Self-Supervised Fusarium Head Blight Detection with Hyperspectral Image and Feature Mining," arXiv:2409.00395, 2024.

Technical Reports and Challenge Papers

1. Z. Chen, K. Liu, Y.-F. Lin, et al., "The Fourth Challenge on Image Super-Resolution (×4) at NTIRE 2026: Benchmark Results and Method Overview," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2026.
2. F.-A. Vasluianu, T. Seizinger, Y.-F. Lin, et al., "Learning-Based Ambient Lighting Normalization: NTIRE 2026 Challenge Results and Findings," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2026.
3. J. Cai, K. Yang, Y.-F. Lin, et al., "NTIRE 2026 Challenge on Single Image Reflection Removal in the Wild: Datasets, Results, and Methods," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2026.
4. Z. Chen, R. Timofte, Y.-F. Lin, et al., "NTIRE 2025 Challenge on Image Super-Resolution (×4): Methods and Results," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2025.
5. Z. Chen, R. Timofte, Y.-F. Lin, et al., "NTIRE 2025 Challenge on Real-World Face Restoration: Methods and Results," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2025.
6. B. Kiefer, L. Žust, Y.-F. Lin, et al., "3rd Workshop on Maritime Computer Vision (MaCVi) 2025: Challenge Results," Technical Report, IEEE WACV, MaCVi Workshop, 2025.
7. F.-A. Vasluianu, R. Timofte, Y.-F. Lin, et al., "NTIRE 2025 Challenge on Single Image Shadow Removal," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2025.
8. K. Yang, R. Timofte, Y.-F. Lin, et al., "NTIRE 2025 Challenge on Single Image Reflection Removal in the Wild: Datasets, Methods and Results," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2025.
9. S. Han, H. Fan, Y.-F. Lin, et al., "NTIRE 2025 Challenge on Text-to-Image Generation Model Quality Assessment," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2025.
10. C.-M. Lee, B.-C. Qiu, Y.-F. Lin, et al., "Multi-Source COVID-19 Detection via Kernel-Density-Based Slice Sampling," Technical Report, IEEE/CVF ICCV, AFEMI Workshop, 2025.
11. Y.-F. Lin, B.-C. Qiu, C.-M. Lee, and C.-C. Hsu, "Divide and Conquer: Grounding Bleeding Areas in Gastrointestinal Image with Two-Stage Model," Technical Report, IEEE ICIP, Auto-WCEBleedGen Challenge V2, 2024.

12. R. Ben, R. Timofte, Y.-F. Lin, et al., "The Ninth NTIRE 2024 Efficient Super-Resolution Challenge Report," Technical Report, IEEE/CVF CVPR, NTIRE Workshop, 2024.
13. Y.-F. Lin, C.-H. Cheng, C.-C. Hsu, and C.-M. Lee, "Simple 2D Convolutional Neural Network-Based Approach for COVID-19 Detection," Technical Report, IEEE ICPR, Hyperspectral Imagery Challenge, 2024.

References

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